Infinite families of rank ≥ 2

Problèmes ouverts — A. Dujella · 2026

Problem 2.3 (Dujella, 2026). Are there infinitely many elliptic curves over $\mathbb{Q}$ with torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ and rank $\geqslant 2$? The same question for torsion groups $\mathbb{Z}/9\mathbb{Z}$, $\mathbb{Z}/10\mathbb{Z}$, and $\mathbb{Z}/12\mathbb{Z}$.

Remark. There are infinitely many curves with rank $\geqslant 1$ and torsion groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ (Atkin & Morain (1993), Kulesz (1998), Lecacheux (2002), Campbell & Goins (2003), Rabarison (2008)), $\mathbb{Z}/9\mathbb{Z}$ (Atkin & Morain (1993), Kulesz (1998), Rabarison (2008), Gasull, Manosa & Xarles (2010), Youmbai (2024)), $\mathbb{Z}/10\mathbb{Z}$ (Atkin & Morain (1993), Kulesz (1998), Rabarison (2008), Voznyy (2021)), and $\mathbb{Z}/12\mathbb{Z}$ (Suyama (1985), Kulesz (1998), Rabarison (2008), Halbeisen, Hungerbühler, Shamsi Zargar & Voznyy (2021)).